CALM connectome figures — style guide

One palette, one type scale, one set of chrome rules, applied to every figure in the project.

Part 1 · the system

Community palette

Okabe-Ito, fixed slot order. Slot k belongs to community k in every figure — never cycled, never re-ranked.

1 orange	2 sky blue	3 bluish green	4 blue	5 vermillion	6 reddish purple	— other
#E69F00	#56B4E9	#009E73	#0072B2	#D55E00	#CC79A7	#898781

Validated all-pairs (any two communities may sit adjacent in a brain render): worst CVD ΔE 7.6 · worst normal-vision ΔE 15.6 · PASS

ΔE 7.6 is in the 6–8 floor band → secondary encoding is MANDATORY: block boundaries, colour bars, direct labels, or facets. Six is a cap, not a default.

Magnitude and polarity

Sequential — one hue, light→dark. Connectivity magnitude.

Diverging — two hues, NEUTRAL grey midpoint. Within − between, group contrasts.

Never a rainbow. Never a hue at the diverging midpoint. Magnitude is never encoded with the community palette — that channel carries identity only.

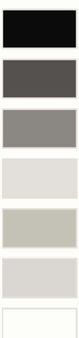
Typography

Lato, installed cluster-wide. Fallbacks: Liberation Sans, DejaVu Sans. Figures authored at 180 mm width.

Figure title	11.0 pt	semibold
Panel title	9.5 pt	semibold
Axis label	8.5 pt	normal
Tick / annotation	7.5 pt	normal
Caption	7.0 pt	normal

Text always wears ink tokens — never a community colour. A coloured mark beside the label carries identity.

Ink & chrome



primary ink	#0b0b0b
secondary ink	#52514e
muted / axis labels	#898781
gridline	#e1e0d9
baseline / axis	#c3c2b7
context node	#d8d7d1
chart surface	#fcfcfb

Marks

- 2.0 pt lines, round caps
- ≥ 4 pt markers (8 px)
- 0.6 pt axes, no top/right spine
- hairline grid, always behind
- 2 px surface ring on overlaps
- legend always ≥ 2 series

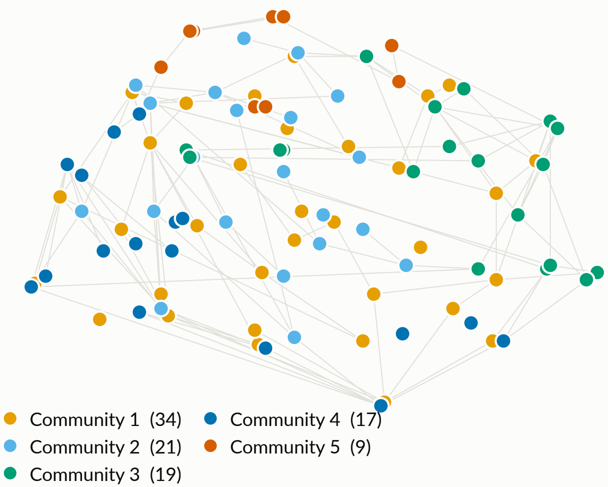
The same six communities, three representations

Group-average CALM 800 connectome (n = 315), Schaefer 100×17, streamline counts. Louvain found 5 communities.

Part 2 · the system applied

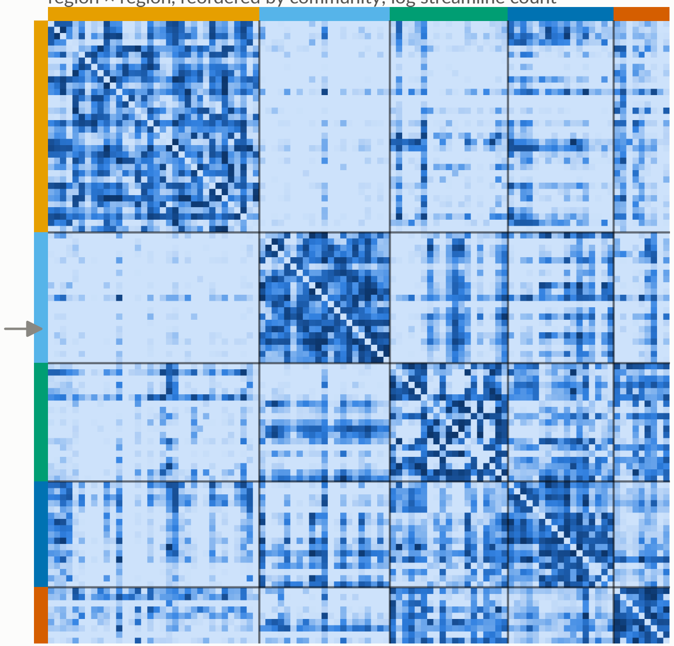
Anatomical space

sagittal view, top 4% of edges



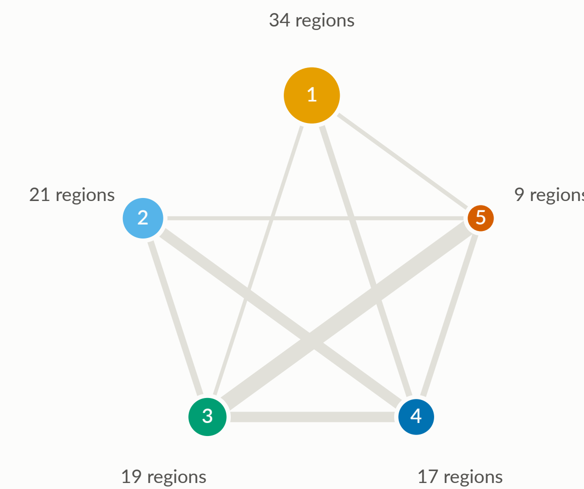
Connectivity space

region × region, reordered by community; log streamline count



Community space

node size = regions; edge width = mean connectivity



Community identity is invariant across the three spaces — that invariance is the whole point of fixing the palette. Every panel carries secondary encoding (block boundaries and colour bars on the matrix, numbered super-nodes on the graph), which the CVD floor band makes mandatory rather than optional.